

Anomaly Detection in Elderly Daily Activities Using Advanced Deep Learning Models

Background and Motivation

With the rapidly aging global population, there is an urgent need for scalable, privacy-preserving solutions to monitor elderly individuals living alone. Traditional monitoring methods (wearables, cameras) often face challenges with user compliance due to their intrusive nature. Electrical load monitoring provides an innovative alternative by analyzing household appliance usage patterns to detect anomalies in Activities of Daily Living (ADL) without compromising privacy.

Problem Statement

Current anomaly detection methods in smart home environments often fail to capture the complex temporal dependencies and multi-scale patterns in human behavior. Standard autoencoder architectures treat power consumption data as independent snapshots, missing crucial sequential information that characterizes normal daily routines.

Proposed Solution

We propose developing a hybrid deep learning architecture that combines Temporal Convolutional Networks (TCN) and Transformer components within a variational autoencoder framework. This model will learn normal behavioral patterns from aggregate household electricity consumption data and detect subtle deviations that may indicate health issues or emergencies.

Datasets

1. UK-DALE (UK Domestic Appliance-Level Electricity)

<https://data.ceda.ac.uk/edc/efficiency/residential/EnergyConsumption/Domestic/UK-DALE-2015>

- Consommation électrique totale + 5 appareils individuels par maison
- 5 foyers britanniques surveillés pendant plusieurs années
- Fréquence d'échantillonnage: 16 kHz (appareils) et 6 secondes (total)

2. REFIT (Electrical Load Measurements)

<https://pureportal.strath.ac.uk/en/datasets/refit-electrical-load-measurements-cleaned>

- 20 foyers écossais surveillés pendant 2 ans
- Consommation totale + 9 appareils principaux par maison
- Fréquence: 8 secondes pour le total, 1-8 secondes pour les appareils

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